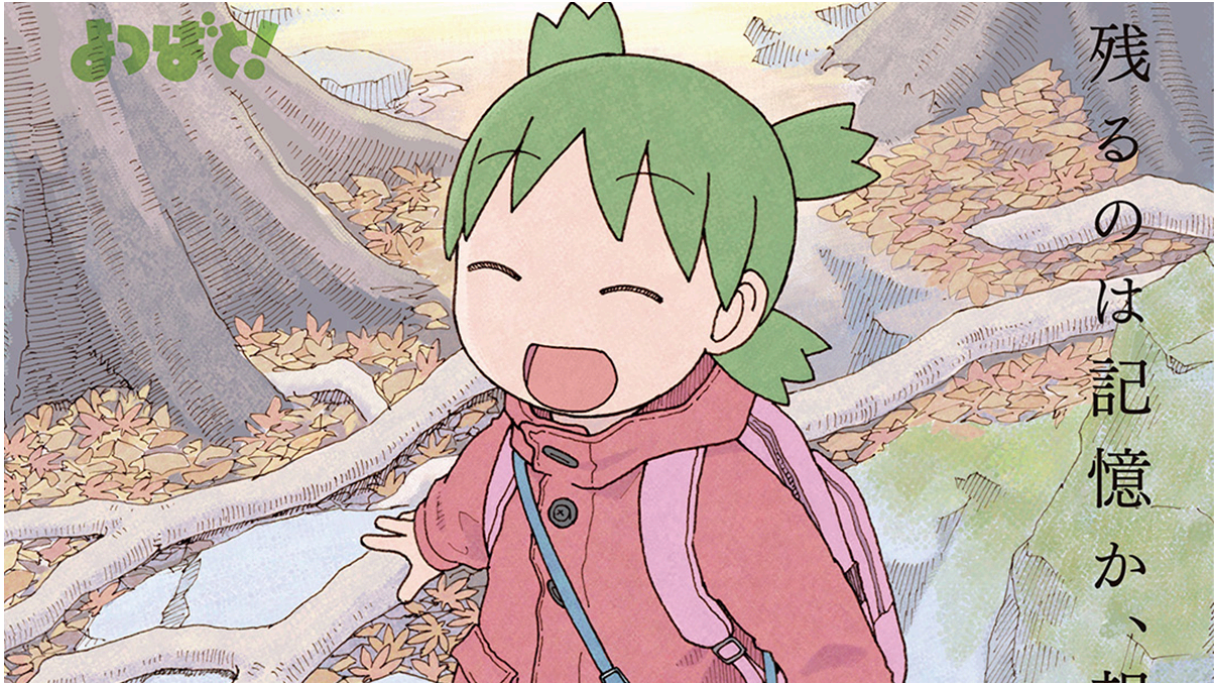


MATH 284 Notes

Linear Algebra

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Yotsuba from Yotsuba&!

Contents

1	Linear Equations in Linear Algebra	3
1.1	Systems of Linear Equations	3
	Matrix Notation	3
	Solving a Linear System	3
1.2	Row Reduction and Echelon Forms	5
	Pivot Positions	5
	The Row Reduction Algorithm	5
	Solutions of Linear Systems	6
1.3	Vector Equations	6
1.4	The Matrix Equation $A\mathbf{x} = \mathbf{b}$	6
1.5	Solution Sets of Linear Systems	6
1.6	Linear Independence	6
1.7	Linear Transformations	6
1.8	Matrices of Linear Transformations	6
2	Matrix Algebra	7
2.1	Matrix Operations	7
2.2	The Inverse of a Matrix	7
2.3	Characteristics of Invertible Matrices	7
3	Determinants	8
4	Vector Spaces	9
5	Eigen-everything	10
5.1	Eigenvectors and Eigenvalues	10
5.2	The Characteristic Equation	11
	Similarity	11
5.3	Diagonalization	13
5.4	Eigenvectors and Linear Transformations	14
5.5	Complex Eigenvalues	14
6	Orthogonality and Least Squares	15
7	Symmetric Matrices and Quadratic Forms	16

1 Linear Equations in Linear Algebra

1.1 Systems of Linear Equations

- A **linear equation** with the variables $x_1, x_2, \dots, x_n$ and coefficients $a_1, a_2, \dots, a_n$ can be written as follows:

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

- A **system of linear equations** is a collection of numerous linear equations all using the same variables.

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = k_1$$

$$b_1x_1 + b_2x_2 + \dots + b_nx_n = k_2$$

- The solutions of a system of linear equations have three variants:
 - **No solution.** In $\mathbb{R}^2$, the linear equations would be parallel, non-overlapping lines.
 - **One solution.** In $\mathbb{R}^2$, the linear equations would be nonparallel lines.
 - **Infinitely many solutions.** In $\mathbb{R}^2$, the linear equations would be parallel, overlapping lines.

If a system of linear equations has a **solution set**, it is **consistent**. Otherwise, it is **inconsistent**.

Matrix Notation

We can display a system of linear equations with a matrix as follows.

$$\begin{array}{rcl} 5x_1 + x_2 - 9x_3 & = & 2 \\ x_1 + 0x_2 + 4x_3 & = & 7 \\ 0x_1 + 0x_2 + 6x_3 & = & 1 \end{array} \quad \begin{pmatrix} 5 & 1 & -9 \\ 1 & 0 & 4 \\ 0 & 0 & 6 \end{pmatrix}$$

- The matrix above is a **coefficient matrix**.
- Each column corresponding to a different variable.
- An **augmented matrix** has an augmented column of the system's constants, usually separated by a line for clarity.

$$\left(\begin{array}{ccc|c} 5 & 1 & -9 & 2 \\ 1 & 0 & 4 & 7 \\ 0 & 0 & 6 & 1 \end{array} \right)$$

Solving a Linear System

To solve a linear system, we can replace it with an equivalent system. We can create an equivalent system by applying an **elementary row operation** to the rows (or in equation form, equations).

Definition 1.1.1 (Elementary Row Operations)

- 1) **Replacement:** Replace one row by the sum of itself and a scalar multiple of another row.
- 2) **Interchange:** Swap two rows.
- 3) **Scaling:** Multiplies all entries of a row by a nonzero constant.



- Thus, two matrices are **row equivalent** if there exists a sequence of elementary row operations that transforms one into the other.
- All elementary row operations are reversible.

1.2 Row Reduction and Echelon Forms

Definition 1.2.1 (Echelon Form)

A matrix is in **echelon form**, also known as **row echelon form**, if it has the following properties:

- 1) All nonzero rows are above any rows of all zeros.
- 2) Each leading (first) entry of a row is in a column to the right of the leading entry of the row above it.
- 3) All entries (in the same column) below a leading entry are zeros.



Definition 1.2.2 (Reduced Row Echelon Form)

Furthermore, a matrix is in **reduced row echelon form (rref)** if it satisfies the previous three properties and the following properties:

- 1) The leading entry in each row is 1.
- 2) Each leading 1 is the only nonzero entry in its column.



Theorem 1.2.3 (Uniqueness of the Reduced Echelon Form)

Each matrix is row equivalent to one reduced echelon form matrix.



- Because the leading values of an echelon form matrix can be anything, they are not unique.

Pivot Positions

- A **pivot position** in a matrix is the location that corresponds to the location of a leading 1 in the matrix's reduced row echelon form.
- A **pivot column** is any column containing a pivot position.

The Row Reduction Algorithm

Algorithm 1.2.4 (Row Reduction Algorithm)

- 1) Begin with the leftmost nonzero column. This will be a pivot column with a pivot position at the top.
- 2) Select a nonzero entry in the pivot column as the pivot. If needed, swap rows to move it into the pivot position.
- 3) Use elementary row operations to turn all entries below the pivot into zeros.
- 4) Repeat steps (1-3) with each consecutive nonzero column with pivot positions one row below the previous one.
- 5) Starting with the rightmost pivot, using scaling, if necessary, to convert each pivot to a 1.



Solutions of Linear Systems

Applying the row reduction algorithm to an augmented matrix leads directly to a solution set for the associated linear system.

- **Basic variables** are variables whose coefficients are in a pivot column.
- **Free variables** are variables whose coefficients aren't in a pivot column.

1.3 Vector Equations

1.4 The Matrix Equation $A\mathbf{x} = \mathbf{b}$

1.5 Solution Sets of Linear Systems

1.6 Linear Independence

1.7 Linear Transformations

1.8 Matrices of Linear Transformations

2 Matrix Algebra

2.1 Matrix Operations

2.2 The Inverse of a Matrix

2.3 Characteristics of Invertible Matrices

Theorem 2.3.1 (Invertible Matrix Theorem)

Let A be an $n \times n$ matrix. The following statements are equivalent:

- a) A is invertible.
- b) A is row equivalent to I_n .
- c) A has n pivot positions.
- d) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- e) The columns of A form a linearly independent set.
- f) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
- g) The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b} \in \mathbb{R}^n$.
- h) The columns of A span $\mathbb{R}^n$.
- i) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- j) There exists some $n \times n$ matrix C such that $CA = I$.
- k) There exists some $n \times n$ matrix D such that $AD = I$.
- l) A^T is invertible.



3 Determinants

4 Vector Spaces

5 Eigen-everything

5.1 Eigenvectors and Eigenvalues

Definition 5.1.1 (Eigenvector and Eigenvalue)

An **eigenvector** of an $n \times n$ matrix A is any nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ . A scalar λ is an **eigenvalue** of A if there exists a nontrivial solution $\mathbf{x}$ to $A\mathbf{x} = \lambda\mathbf{x}$ such that $\mathbf{x}$ is the eigenvector corresponding to λ .

- An **eigenspace** consists of all eigenvectors corresponding to some eigenvalue λ , and forms a subspace of $\mathbb{R}^n$.
- Geometrically, an eigenvalue is a scaling factor in a specific direction during a linear transformation.

Note

While eigenvectors cannot be zero, *eigenvalues can be zero*.

5.2 The Characteristic Equation

To solve for the eigenvectors, and by extension, the eigenvalues of a matrix, we can take advantage of the general matrix equation for eigenvectors.

$$\begin{aligned} A\mathbf{x} &= \lambda\mathbf{x} \\ A\mathbf{x} - \lambda\mathbf{x} &= 0 \\ (A - \lambda I_n)\mathbf{x} &= 0 \end{aligned}$$

- Now, by definition, the homogeneous equation above must have a nontrivial solution.
- Then, by the Invertible Matrix Theorem, $(A - \lambda I_n)$ is not invertible. Equivalently, its determinant is zero. This gives us the **characteristic equation**, a n^{th} degree polynomial whose roots are A 's eigenvalues.

Definition 5.2.1 (Characteristic Equation)

Let A be some $n \times n$ matrix. The characteristic equation is defined as follows:

$$\det(A - \lambda I_n) = 0$$

A scalar λ may only be an eigenvalue for A if it satisfies the equation above.

- We can use this to prove that for any $n \times n$ matrix A , A^T has the same eigenvalues.

Proof. Suppose A and A^T are some $n \times n$ matrices whose eigenvalues are given by $\det(A - \lambda I)$ and $\det(A^T - \lambda I)$, respectively.

$$\begin{aligned} \det(A^T - \lambda I) &= \det(A^T - \lambda I^T) \\ \det(A^T - \lambda I) &= \det(A - \lambda I) \end{aligned}$$

□

Similarity

If A and B are $n \times n$ matrices, then A is **similar to** B if there exists an invertible matrix P such that $A = PBP^{-1}$, or equivalently, $B = P^{-1}AP$.

- This relationship is *symmetric*.
- The mapping $A \mapsto P^{-1}AP$ is known as a **similarity transformation**.

Theorem 5.2.2

If two $n \times n$ matrices A and B are similar, then their characteristic polynomials, and by extension, their eigenvalues, are the same.

Proof. Let P be some $n \times n$ invertible matrix.

$$A - \lambda I = PBP^{-1} - \lambda PP^{-1}$$

$$A - \lambda I = P(B - \lambda)P^{-1}$$

$$\det(A - \lambda I) = \det(P(B - \lambda)P^{-1})$$

$$\det(A - \lambda I) = \det(P(B - \lambda)P^{-1})$$

$$\det(A - \lambda I) = \det(P) \det(B - \lambda) \det(P^{-1})$$

$$\det(A - \lambda I) = \det(P) \det(B - \lambda) \left(\frac{1}{\det(P)} \right)$$

$$\det(A - \lambda I) = \det(B - \lambda)$$

□

5.3 Diagonalization

A matrix is **diagonalizable** if it is similar to a diagonal matrix.

- This factorization is particularly helpful for calculating large k^{th} powers of matrices.

Theorem 5.3.1 (The Diagonalization Theorem)

An $n \times n$ matrix A is diagonalizable $\Leftrightarrow A$ has n linearly independent eigenvectors.

Specifically, $A = PDP^{-1}$ given diagonal matrix $D \Leftrightarrow$ the columns of P are n linearly independent eigenvectors of A and the diagonal entries of D are the eigenvalues corresponding to each i^{th} column in P .

- In other words, A is diagonalizable $\Leftrightarrow$ there exists enough eigenvectors to form an **eigenvector basis** of $\mathbb{R}^n$.

Proof. Let P be the a matrix whose columns are A 's eigenvectors, $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and let D be a diagonal matrix whose diagonal entries are A 's eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Since the eigenvectors of A are linearly independent, the columns of P are linearly independent. By the Invertible Matrix Theorem, P is invertible. Since P , D , and P^{-1} are all well-defined, the diagonalization PDP^{-1} exists as follows:

$$\begin{aligned} AP &= A(\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n) \\ AP &= (A\mathbf{v}_1 \ A\mathbf{v}_2 \ \cdots \ A\mathbf{v}_n) \\ AP &= (\lambda_1\mathbf{v}_1 \ \lambda_2\mathbf{v}_2 \ \cdots \ \lambda_n\mathbf{v}_n) \\ APP^{-1} &= (\lambda_1\mathbf{v}_1 \ \lambda_2\mathbf{v}_2 \ \cdots \ \lambda_n\mathbf{v}_n)P^{-1} \\ A &= (\lambda_1\mathbf{v}_1 \ \lambda_2\mathbf{v}_2 \ \cdots \ \lambda_n\mathbf{v}_n)P^{-1} \\ A &= PDP^{-1} \end{aligned}$$

Now, suppose A is diagonalizable such that $A = PDP^{-1}$. This is equivalent to $AP = PD$. Now, let P 's columns be vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and let D 's diagonal entries be $\lambda_1, \lambda_2, \dots, \lambda_n$.

$$AP = PD$$

$$\begin{aligned} A(\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n) &= (\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n) \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \\ (A\mathbf{v}_1 \ A\mathbf{v}_2 \ \cdots \ A\mathbf{v}_n) &= (\lambda_1\mathbf{v}_1 \ \lambda_2\mathbf{v}_2 \ \cdots \ \lambda_n\mathbf{v}_n) \end{aligned}$$

□

5.4 Eigenvectors and Linear Transformations

5.5 Complex Eigenvalues

6 Orthogonality and Least Squares

Least

7 Symmetric Matrices and Quadratic Forms